

[Total No. of Questions - 9] [Total No. of Printed Pages - 4]
Dec-24-0341 (NEP)

MCA-7301 (Operational Research)
MCA-3rd CBCS/NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section- A

1. (a) Define linear programming. Discuss the assumptions of linear programming model. (4)
(b) Solve by using simplex method. (8)
Maximize $Z = 4x_1 + 10x_2$
Subject to
 $2x_1 + x_2 \leq 10$
 $2x_1 + 5x_2 \leq 20$
 $2x_1 + 3x_2 \leq 18$
 $x_1, x_2 \geq 0$
2. (a) What do you mean by degeneracy in LPP? Explain with the help of suitable example. (4)
(b) Solve by Two Phase simplex method.
Maximize $8x_1 + 5x_2$
Subject to :
 $3x_1 + 2x_2 \geq 3$
 $x_1 + 4x_2 \geq 4$
 $x_1 + x_2 \leq 5$
 $x_1, x_2 \geq 0$ (8)

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Section- B

3. An architect has been awarded a contract to prepare plans for an urban renewal project. The job consists of the following activities and their estimated times

Activity	Description	Immediate predecessors	Time (Days)
A	Prepare preliminary sketches	-	2
B	Outline specifications	-	1
C	Prepare drawings	A	3
D	Write specifications	A,B	2
E	Run off prints	C,D	1
F	Have specification	B,D	3
G	Assemble bid packages	E,F	1

- (a) Draw the network diagram of activities for the project.
 - (b) Indicate the critical path and calculate the total float and free float for each activity. (12)
4. Find the basic feasible solution to the following transportation problem by using:
(a) North-West Corner Rule

[P.T.O.]

(b) Vogel's approximation method

Distribution Centers					
Sources	D1	D2	D3	D4	Supply
S1	2	3	11	7	6
S2	1	0	6	1	1
S3	5	8	15	9	10
Requirements	7	5	3	2	

(12)

Section -C

5. In a game of matching coins with two players, suppose A wins one unit of value when there are two tails and loses 1/2 unit of value when there is one head and one tail. Determine the pay-off matrix, the best strategies for each player and the value of the game to A. (12)
6. A self-service store employs one cashier at its counter. An average of nine customers arrives every 5 minutes, while the cashier can serve 10 customers in 5 minutes. Assuming Poisson distribution for arrival rate and exponential distribution for service rate, find :
 - (a) The average number of customers in the system.
 - (b) The average number of customers in a queue or average queue length.
 - (c) Average time a customer spends in the system.
 - (d) Average time a customer waits before being served. (12)

Section-D

7. (a) The demand for an item is 18000 units per year. The holding cost is Rs 1.20 per unit time and the cost of shortage is Rs.5.00. The production cost is Rs.400.00. Assuming that replacement rate is instantaneous determine the optimum order quantity. (8)
- (b) What do you mean by buffer stock? Explain (4)
8. What do you mean by inventory management? Discuss the various types of inventory and also discuss the importance of inventory management in detail. (12)

Section-E (Compulsory)

9. a. Discuss limitations of LPP.
- b. What do you mean by duality in LPP?
- c. What do you mean by optimistic, most likely and pessimistic time in PERT?
- d. Define EOQ.
- e. Explain LIFO, FIFO systems.
- f. With the help of suitable example explain two- person zero-sum game. (6×2=12)